

Leveraging LLMs for Environmental Complexity: Structured Fine-Tuning Data Sets and Deployment Strategies

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ABSTRACT: Generative artificial intelligence, especially large language models (LLMs), could accelerate environmental analysis, but deployment is hindered by two gaps: limited structured domain knowledge and unclear strategies matched to environmental decision contexts. Here, this study constructs a textbook-based, China-centered environmental knowledge data set with hierarchical organization to enable reliable fine-tuning and benchmarking. Results show a consistent trade-off that fine-tuned models achieve modest gains in precision (+1%) and response efficiency (+52%) on standardized tasks but exhibit limited adaptability when embedded in agentic workflows (−3%). In contrast, state-of-the-art generalist models consistently outperform in system-level sustainability and interdisciplinary decision tasks (+10%), benefiting from stronger cross-domain reasoning and dynamic tool integration. Together, these findings support a layered LLMs' deployment strategy for environmental intelligence. Specifically, selective fine-tuning for stable, regulatory, and verification tasks, combined with agentic workflows anchored in up-to-date generalist backbone models for dynamic, data-intensive, and interdisciplinary decision-making. This work provides both a reusable data set foundation and a practical framework for deploying LLMs as scalable and reliable decision-support tools in environmental decision.

KEYWORDS: environmental complexity, generative artificial intelligence, large language models, fine-tuning, agentic workflows

1. INTRODUCTION

Generative artificial intelligence (GAI), especially for large language models (LLMs), has emerged as a powerful tool for addressing critical challenges in environmental domains. However, these challenges are inherently complex due to their interdisciplinary nature, global interdependencies, and systemic uncertainties.¹ Environmental problems often span multiple domains² while integrating heterogeneous data sources (e.g., satellite imagery, sensor networks, and textual reports) with varying spatiotemporal scales. Additionally, incomplete knowledge, dynamic feedback loops, and conflicting stakeholder priorities further complicate solutions.^{3,4}

Despite these barriers, recent studies demonstrate its versatility in environmental applications, from predicting catalytic reaction outcomes (e.g., NO_x reduction with NH₃) using structured reasoning frameworks⁵ to enhancing quantitative structure–activity relationship (QSAR) models for water contaminant properties,⁶ outperforming conventional approaches and even human experts. LLMs have also enabled the automated extraction of toxicological data, facilitating the identification of pollutants and their oxidative stress mechanisms at an unprecedented scale.⁷ Metal–organic framework

(MOF) synthesis has been predicted using the similar strategy by incorporating chemistry-specific prompts with ChatGPT.⁸ The use of LLMs to automate quality assurance and control screening for microplastics research shows that AI-assisted evaluation can significantly improve data consistency and reliability for environmental risk assessment.⁹ Notably, AI-driven emission factor recommendation systems have streamlined life cycle assessments (LCA), improving the accuracy and efficiency of quantifying greenhouse gas emissions.¹⁰ Also, AI automates emission inventory determination for product carbon footprints, accelerating assessments,¹¹ significantly improving assessment efficiency. Specialized models such as AMR-Policy GPT¹² leverage multilingual policy databases to support evidence-based antimicrobial resistance strategies, particularly in resource-limited settings. These applications

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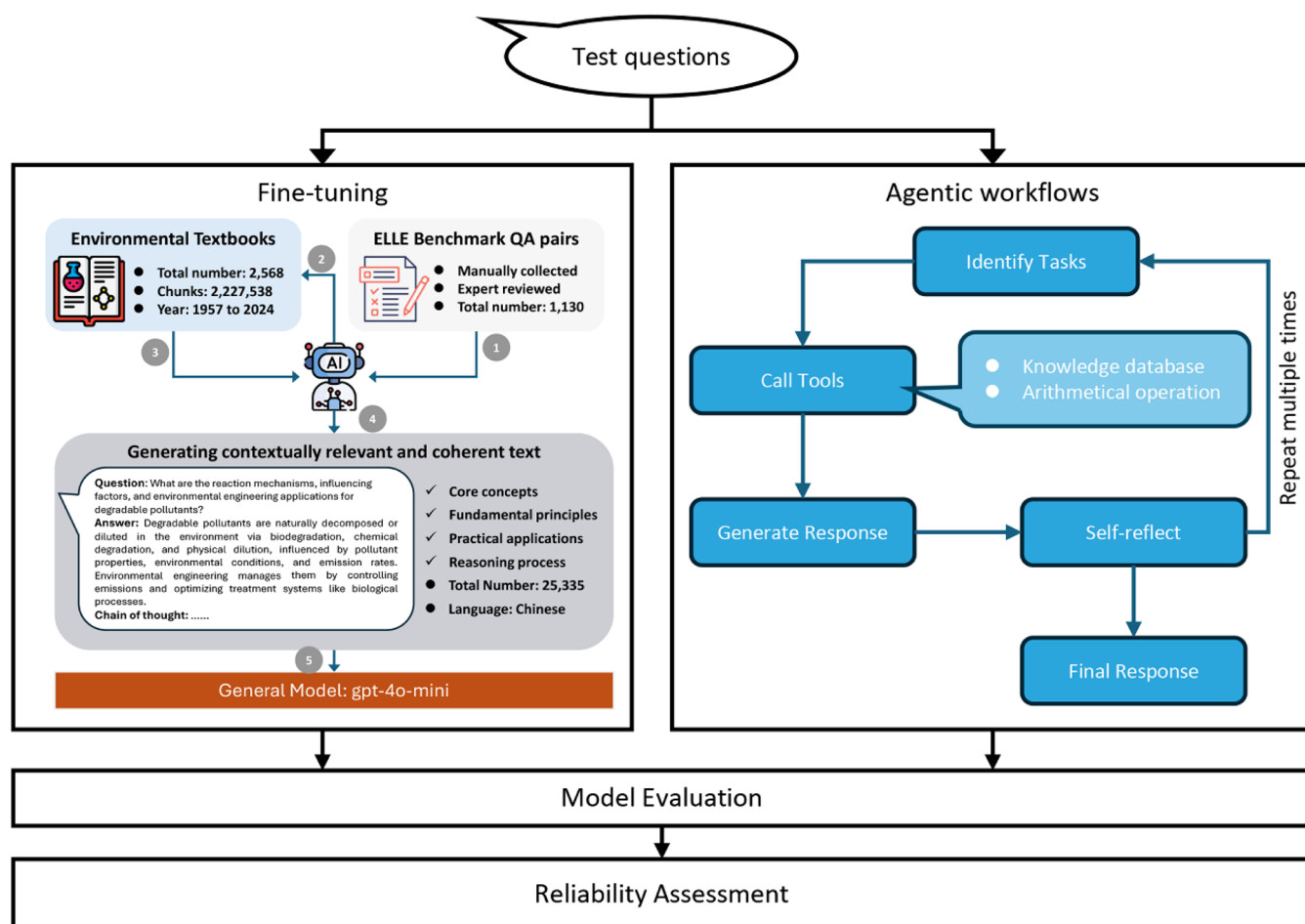


Figure 1. Architecture of textbook-based fine-tuning pipeline (left), agentic workflow with tools and reflection loop (right), model performance evaluation, and reliability assessment.

showcase the potential of LLMs to revolutionize environmental monitoring, pollution control, and climate mitigation through data-driven automation and enhanced decision-making.

Despite their potential, LLMs face critical limitations in environmental applications. A major challenge is their tendency to produce hallucinations (i.e., factually incorrect or unverified outputs), which undermines reliability in high-stakes domains like pollutant identification or emission modeling. As a result, their responses can exhibit inconsistency even for identical queries (e.g., identifying pH tolerance ranges or catalytic conditions), introducing uncertainty in reproducibility-sensitive tasks. Additionally, LLMs often rely on outdated information due to static training data sets, limiting their ability to adapt to evolving environmental regulations or emerging contaminants. These issues, compounded by the lack of robust validation frameworks, highlight the need for improved model stability, real-time knowledge integration, and hybrid approaches combining domain expertise to ensure trustworthy environmental decision-making.

Fine-tuning is one of the gold standards for addressing these issues.^{13–15} Recent benchmarking studies (e.g., WaterGPT,¹⁶ OceanGPT,¹⁷ EnvGPT¹⁸) demonstrate fine-tuning's superiority over prompt engineering for specialized tasks like water knowledge applications. However, its adoption faces a critical barrier: the scarcity of high-quality, domain-specific training data.^{19,20} Prior attempts merely shift these requirements rather

than solve them, for example, constructing a language model—CLIMATEBERT that is specifically pretrained on climate-related text corpora²¹ and developing a hybrid AI system (i.e., ChatClimate²⁰) that combines the internal knowledge of foundation models with authoritative external knowledge, demonstrating improved accuracy. These methods still demand an authoritative reference corpus or massive in-domain text collections (e.g., over 2 million climate-related paragraphs for CLIMATEBERT²¹). Either curated, high-quality reference corpora or large-scale in-domain text collections are unavailable in many specialized domains. The critical challenge of acquiring domain-specific training data has spurred innovative approaches to data set construction. Notably, textbook-based knowledge has emerged as a high-value resource due to its unique advantages,²² as its content is pedagogically sequenced for gradual knowledge building and inherently reliable, as typically peer-reviewed or institutionally endorsed. The significance of textbook-quality data is exemplified,²³ where training on 6B tokens of curated web-based textbooks and 1B tokens of synthetic textbook exercises enabled a 1.3 B-parameter model to achieve higher accuracy, demonstrating that textbook-derived data can compensate for small model scales through quality.

Besides, agentic workflows have emerged as a promising approach to complex knowledge tasks, where LLMs operate as autonomous agents capable of iterative reasoning, tool use, and multistep problem-solving.^{24–26} Recent explorations have

demonstrated their potential in domains such as software engineering, healthcare, and life sciences (e.g., drug discovery²⁷), robotics and automation,^{28,29} and scientific literature review (e.g., AI agents synthesizing research findings).^{30,31} However, their application in environmental science remains underexplored, with limited studies systematically evaluating their efficacy against fine-tuned models for domain-specific tasks.

This study investigates how LLMs can be effectively adapted and deployed for environmental science and engineering within a China-centered knowledge framework. It addresses two critical gaps: the data gap, referring to the limited availability of high-quality, domain-specific training resources, and the knowledge gap, concerning the absence of systematic approaches for integrating LLMs into environmental analytical workflows. To address these challenges, we developed a textbook-based data set that hierarchically organizes environmental knowledge from Chinese-authored materials, providing a structured and interpretable representation of disciplinary concepts. Using this data set, we fine-tuned a domain-specific model and benchmarked its performance against general-purpose and agentic architectures to examine how specialization, generalization, and tool integration jointly affect model adaptability. The resulting analysis provides empirical evidence for when targeted fine-tuning is beneficial and when generalist or agentic approaches offer superior flexibility, thereby informing the design of future environmental LLMs and guiding their application in region-specific yet globally relevant contexts. In this sense, the contribution of this work lies not in model optimization alone, but in articulating a deployment logic that aligns AI capabilities with the demands of environmental decision-making.

2. MATERIALS AND METHODS

This study develops a China-centered specialized environmental LLM through a systematic approach involving textbook-based knowledge integration, high-quality training data synthesis, and fine-tuning. Additionally, we design an agentic workflow that equips the LLM with the capability to utilize external tools, such as retrieval-augmented generation (RAG) and arithmetic operations, while enabling self-evaluation and reflective response refinement. To assess performance, we conduct comparative evaluations among fine-tuned models, base models, state-of-the-art (SOTA) models, and agentic benchmarking workflows using these three models. An overview of these processes is illustrated in Figure 1, with detailed materials and methods described in the following subsections.

2.1. Train Data Preparation

2.1.1. Textbook Corpus Compilation. We systematically compiled a comprehensive collection of 2,568 authoritative environmental science textbooks, spanning 1957 to 2024 to ensure coverage of both foundational theories and the latest advancements. As shown in Supporting Information (Section 1), these textbooks encompass 10 major domains, including environmental economics, environmental management, environmental chemistry, and environmental physics, ensuring broad thematic representation. While the collection is global in scope, the majority (around 95%) are Chinese-authored textbooks, making it particularly valuable for analyzing China-specific environmental policies, case studies, and regulatory frameworks. All textbook-derived materials used in

this study were accessed through the university's licensed digital repository and processed solely for noncommercial academic research. No verbatim text or copyrighted material is redistributed. The derived data set consists of paraphrased, agent-generated question–answer pairs created through automated knowledge abstraction. Source-processing scripts and extraction workflows are openly available to ensure methodological transparency and compliance with institutional and copyright policies.

For more effective knowledge retrieval, we implemented a dual-database system: (1) A vector database created by chunking texts (total number: 2,227,538) using the optical character recognition method, embedding via text-embedding-3-small model (1536 dimensions), and indexing in Pinecone with cosine similarity metric; (2) A parallel full-text database in OpenSearch, allowing keyword searching. The entire preprocessing pipeline, including chunking algorithms and embedding processes, has been documented and open-sourced to guarantee full reproducibility.³²

2.1.2. Training Data Acquisition. Building upon our previously developed Environmental Large Language Model Evaluation (ELLE) data set, comprising 1,130 expertly validated question–answer pairs across various environmental domains, such as ecology, soil, environmental chemistry, and environmental economics.³³ Then, the training data acquisition process employs an LLM-based agentic data-acquisition workflow to automatically extract high-quality, textbook-derived knowledge. The prompt schema, API parameters, and implementation code are openly available,³⁴ ensuring methodological transparency and independent reproducibility.

This agentic workflow begins with a hybrid retrieval module that queries both vector and full-text databases using seed questions drawn from the ELLE benchmark. For each query, the system retrieves semantically relevant textbook passages and their extended contexts, followed by deduplication to remove redundant content. The retrieved texts are then analyzed by gpt-4o agents operating under a rigorously defined academic protocol. Within each iteration, the agent performs three core steps: formulation of key factors, mechanisms, and solutions as structured academic questions; generation of comprehensive, textbook-grounded responses that integrate theoretical principles, analytical methods, numerical calculations, and real-world applications; and production of detailed reasoning chains documenting the problem decomposition, logical inference, and supporting evidence. To ensure consistency and factual precision, the workflow enforces schema-validated outputs and deterministic decoding (temperature = 0), which substantially mitigates hallucination propagation. This process transforms raw textbook content into a structured and traceable training corpus containing question–answer pairs enriched with reasoning traces and pedagogical distractors, suitable for reinforcement learning from AI feedback.

2.1.3. Training Data Validation. To ensure factual grounding, each AI-generated Q&A entry was reviewed in a structured format containing the question, model response, a reasoning trace (used only to aid expert inspection), and the retrieved textbook chunk from the RAG pipeline. Domain experts compared each response directly with the cited text and assessed factual accuracy, conceptual correctness, and relevance. Entries (~1%, 207) showing unsupported claims, hallucinated numerical values, or mismatches between the question and the retrieved content were removed. The

reasoning traces were not included in the final fine-tuning data. This expert-in-the-loop process yielded a curated data set of 19,532 textbook-aligned and expert-approved Q&A pairs for model training. Summary statistics of retained and discarded entries, the validation workflow, and representative examples of expert decisions are provided in [Supporting Information \(Section 2\)](#). Processed data sets for this agentic pipeline are publicly accessible.³⁵

2.2. Model Fine-Tuning

We fine-tuned the foundation model (i.e., gpt-4o-mini-2024-07-18, hereafter gpt-4o-mini) using supervised fine-tuning (SFT) on 19,532 curated question–answer pairs (approximately 22 million tokens) derived from environmental science textbooks. Training was performed via OpenAI's API under consistent settings to ensure cross-model comparability. Hyperparameters were set to two batches, three epochs, and a learning-rate multiplier of 0.2, based on pilot experiments suggesting stable convergence and validation accuracy. These values fall within empirically supported ranges for small-parameter (<8 B) LLMs. Prior studies^{36–39} indicate that small batch sizes yield stable, hyperparameter-robust optimization, while lower learning rates paired with moderate batches enhance generalization in small-model regimes. Theoretical analyses further show that for Adam-type optimizers (that use adaptive gradient methods of LLM fine-tuning), optimal learning rates decrease with smaller batch sizes,⁴⁰ supporting our configuration.

2.3. Agentic Benchmarking Workflow Construction

This agentic workflow was developed for the experimental comparison of models, which functions as an adaptive problem-solving framework, combining task classification, retrieval-assisted reasoning, numerical computation, and iterative self-reflection. Unlike the data-acquisition workflow, which transforms text into training data, the benchmarking workflow actively engages with test questions to produce grounded, evidence-supported answers. Results and discussions in later sections refer exclusively to this benchmarking workflow.

The workflow^{41,42} adopts a controller–subgraph architecture that enables dynamic task routing and tool use. A top-level controller first classifies each query into calculation, reasoning, or factual tasks. Calculation tasks trigger a specialized subgraph that retrieves relevant equations and standards from environmental databases and then executes Python-based computations to produce intermediate results and contextualized explanations. Reasoning tasks follow a retrieval–generation loop focused on conceptual analysis, while factual queries are answered through lightweight direct generation without tool invocation. Retrieved passages and computed outputs are fed back into the model context, allowing subsequent steps to synthesize numerical and textual evidence into coherent answers.

After each round, a self-reflection module scores the draft response (0–30) on coherence, numerical correctness, and evidence consistency. Scores below 25 or detected inconsistencies reenter the decision loop, prompting targeted revision. According to prior experiments, each cycle typically involves no more than five model invocations before convergence. This reflective cycle typically converges within five iterations. All compared agentic variants share identical workflow logic. Only the underlying backbone model differs.

This agentic workflow architecture was selected to address two persistent challenges in environmental modeling: the need for interpretable, evidence-grounded reasoning rather than opaque text generation, and the requirement to integrate symbolic computation and data retrieval with contextual understanding. The controller–subgraph design enables modular execution and traceable decision flow, aligning with recent best practices in scientific AI systems such as ReAct and LangGraph-based multiagent frameworks.⁴³ The inclusion of a self-reflection loop is empirically motivated by its demonstrated ability to reduce variance and logical inconsistency in iterative reasoning tasks, as reported in adaptive agent studies (e.g., Sparks,⁴⁴ ProtAgents⁴⁵). RAG was incorporated to ensure factual grounding in authoritative environmental data sources, a critical requirement in sustainability and regulatory domains.¹⁸ Together, these components were empirically validated in pilot experiments, where self-reflective iterations reduced variance relative to single-pass baselines,^{46,47} supporting their adoption for final benchmarking.

2.4. Model Evaluation

To ensure a rigorous and unbiased evaluation, we constructed a 328-question test set designed to probe diverse environmental reasoning skills. The questions were drawn from expert-authored items and national examination materials and span multiple cognitive levels, environmental domains, reasoning types, and functional capabilities, enabling assessment of factual grounding, mechanistic understanding, quantitative problem-solving, and cross-domain analytical abilities. Each question includes a verified reference answer to ensure objective scoring. The test set is strictly disjointed from the textbook-derived training data, ensuring that performance reflects genuine generalization rather than memorization. Detailed category definitions and distributions are provided in the [Supporting Information Section 3](#).

We evaluated the base model (gpt-4o-mini), its fine-tuned variant (gpt-4o-mini-ft), a state-of-the-art comparator (gpt-4.1 mini-2025-04-14), and their corresponding agentic workflows. In this study, “agentic workflows” refer strictly to the benchmarking workflow used during testing and are distinct from the data-construction agents used in fine-tuning. All backbones operate within a similar parameter scale (~7–8B^{48–50}), minimizing confounding from model size. Gpt-4.1 mini was selected as the SOTA comparator due to architectural improvements—enhanced attention efficiency, extended context length (1 M versus 128k tokens), and refined distillation mechanisms, representing qualitative design gains rather than scaling effects.^{51,52}

Model responses were scored using a two-stage hybrid protocol. First, LLM evaluators independently assigned blinded 0–100 scores using a standardized five-tier rubric assessing factual correctness, logical coherence, and completeness of quantitative reasoning, with conciseness incorporated under clarity. Each evaluator produced a short rationale for its score. Second, human experts reviewed these rationales for rubric alignment; no inconsistencies were found. Final scores therefore represent expert-validated judgments. This AI-assisted but expert-confirmed framework ensures transparency and reproducibility. All scoring prompts, structured output formats, and executable code are provided for independent verification ([Supporting Information Section 4](#)).

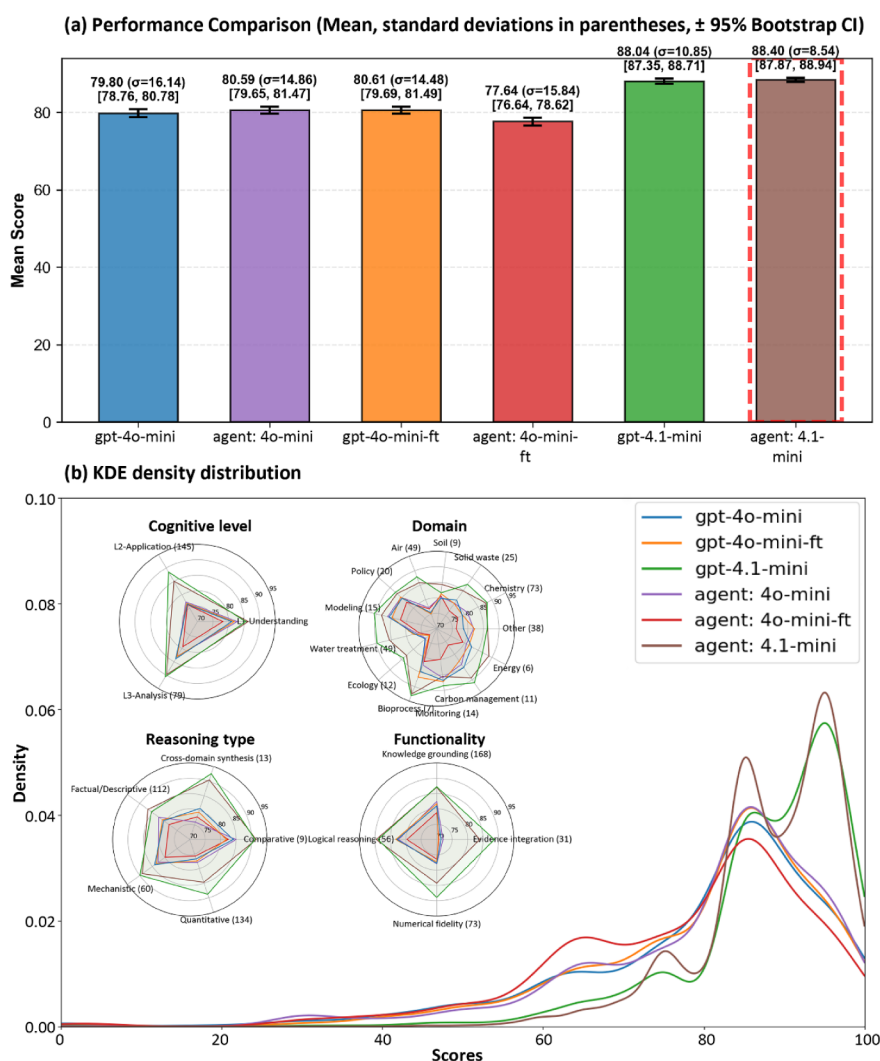


Figure 2. Performance comparison among the fine-tuned, baseline, and SOTA models and their agentic workflows. (a) Mean scores ($\pm$ SD) with 95% bootstrap confidence intervals across six model configurations; (b) Kernel density estimation (KDE) distributions illustrating score dispersion and stability; and scores breakdown across cognitive levels, domains, reasoning types, and functional categories, highlighting how fine-tuning, agentic workflows, and generalist architectures diverge in their strengths across different task dimensions.

2.5. Reliability Assessment

Prior to conducting the multirater analysis, a prescreening step was performed to identify outlier evaluators with insufficient pairwise agreement (Cohen's Kappa < 0.4). This criterion-based exclusion was established a priori to prevent low-consistency raters from distorting the aggregated reliability metrics and to ensure the validity of the multirater assessment. To rigorously assess agreement patterns among the five LLM evaluators (gpt-4.1, DeepSeek-V3, Qwen2.5, Gemini-2.0-flash, and Gemini-2.5-pro), we implemented a sequential analytical approach combining pairwise and multirater agreement metrics through analyses on Cohen's kappa and Fleiss' kappa.

- (1) **Cohen's Kappa Calculation:** We began by computing Cohen's kappa (κ_c) for all possible evaluator pairs (10 unique pairs) to identify systematic disagreements between specific models, detect potential outlier evaluators that might skew the overall agreement assessment, and thus validate the appropriateness of subsequent Fleiss' kappa (κ_f) analysis. Before calculation, we converted the continuous scores into a 5-point ordinal scale (referred to as category j) using predefined

bins [0, 20, 40, 60, 80, 100]. This discretization preserved meaningful distinctions in response quality while ensuring computational stability. P_o (observed agreement) is the proportion of items where two raters (r_1 and r_2) assigned identical scores, see eq 1, where n is the total number of evaluated items (328). P_e (expected chance agreement) estimates the probability of two raters agreeing by random chance, calculated from the marginal distributions of their categorical ratings. For a pair of raters (eq 2), P_j is the proportion of ratings assigned to category j by one rater, while q_j is the analogous proportion for the other rater. k is the number of categories, which is 5 in this study. The κ_c (eq 3) quantifies agreement beyond chance, where P_o is the observed agreement and P_e corrects for random consensus. $\kappa_c > 0$ indicates better-than-chance agreement, with higher values ($\kappa_c > 0.4$) reflecting meaningful reliability.⁵³

$$P_o = \frac{\sum_{i=1}^n I(r_{1,i} = r_{2,i})}{n} \quad (1)$$

$$P_e = \sum_{j=1}^k (p_j \times q_j) \quad (2)$$

$$\kappa_c = (P_o - P_e) / (1 - P_e) \quad (3)$$

- (2) **Fleiss' Kappa Calculation:** Following the exclusion of raters demonstrating poor pairwise agreement ($\kappa_c < 0.4$), we conducted a comprehensive Fleiss' kappa analysis to evaluate inter-rater reliability among the remaining raters. For each of the 384 test items, we constructed an $n \times k$ agreement matrix (where n = number of items, $k=5$ scoring categories) documenting the distribution of ratings across the ordinal scale. Each cell c_{ij} in this matrix records how many of the R retained raters assigned item i to category j ($j = 1, 2, 3, 4$, and 5).

The Fleiss' kappa value is then calculated as eq 8, where $\bar{P}$ represents the observed agreement among raters and P_e accounts for agreement expected by chance. Specifically, $\bar{P}$ was computed by squaring and summing the counts in each category ($\sum_j c_{ij}^2$) across all items (eq 4), adjusting for the number of raters (subtracting m and dividing by $r(r-1)$, eq 5), and averaging these values across all items (multiplying by $1/n$). The chance agreement P_e (eq 7) was derived by first calculating each category's overall proportion (p_j , eq 6), then summing the squares of these proportions. This approach yields a normalized measure where $\kappa_f = 1$ indicates perfect agreement, $\kappa_f = 0$ reflects agreement equal to chance, and negative values suggest systematic disagreement. Based on established benchmarks, we interpreted $\kappa_f > 0.4$ as indicating agreement between evaluators.

This methodology ensured the reliability assessment focused only on consistent evaluators while properly accounting for both observed consensus and chance agreement in the multirater ordinal scoring system. The Fleiss' kappa implementation provided a conservative estimate of scoring reliability for analyses on models' performance.

$$P_i = \frac{\sum_j c_{ij}^2 - r}{r(r-1)} \quad (4)$$

$$\bar{P} = \frac{1}{n} \sum_{i=1}^n P_i \quad (5)$$

$$p_j = \frac{\text{Total ratings in category } j}{k \times n} \quad (6)$$

$$P_e = \sum_j p_j^2 \quad (7)$$

$$\kappa_f = (\bar{P} - P_e) / (1 - P_e) \quad (8)$$

- (3) **Statistical Validation:** A streamlined statistical validation tries to identify that performance differences across models were both reliable and meaningful. Pairwise comparisons using the Benjamini–Hochberg False Discovery Rate (FDR) corrected paired t tests and Wilcoxon tests showed consistent significance, while 10,000-sample bootstrap confidence intervals indicate stable mean estimates with partially nonoverlapping

ranges for higher-performing models. Complementary effect-size analyses (Cohen's d) further indicated that these differences, though sometimes modest in magnitude, represent practical performance gains beyond statistical significance. Statistical methodologies and comparison results are detailed in [Supporting Information \(Section 5\)](#).

3. RESULTS

3.1. Fine-Tuning Improves Performance and Efficiency

Our experimental results (Figure 2) indicate that supervised fine-tuning leads to modest but consistent within-benchmark performance differences. When evaluated on 328 environmental questions, the gpt-4o-mini-ft model shows a gain in average score from 79.8 to 80.2 with reduced variance (from 13.7 to 12.6), indicating improvement and greater stability. Gaussian KDE analysis further clarifies the nature of this improvement. Fine-tuning does not shift the primary performance mode, but it suppresses the lower-score tail and concentrates probability mass around the central region, as reflected by a 7% contraction of the 90% mass interval, an upward shift of the fifth percentile, and a higher modal density despite unchanged peak location. These distributional changes indicate that the fine-tuned model is less prone to low-quality outputs and exhibits more consistent performance, even though its peak capability remains comparable to the base model. Statistical validation supports that the fine-tuned model achieves a mean of 80.2 with a narrower confidence interval ([79.2, 81.2]) relative to the base model (79.8; [78.8, 80.8]), indicating reduced variability and more consistent performance relative to the base model.

Beyond aggregate metrics, five qualitative comparison cases ([Supporting Information, Section 6](#)) illustrate observable differences in explanation structure. These examples show that the fine-tuned model exhibits superior domain grounding, causal reasoning, and structural organization, particularly in explaining mechanistic processes (e.g., smoky smoke flow) and performing structured comparative analyses (e.g., bag filters vs electrostatic precipitators). In contrast, generalist models tend to produce fluent but surface-level narratives, while agentic variants sometimes overexpand contextual details at the expense of technical precision. Together, these results indicate that fine-tuning enhances not only accuracy and efficiency (mean score of 80.6 vs 79.8, and latency of 5.3 s vs 9.0 s), but also the mechanistic completeness and interpretive depth of environmental reasoning tasks.

Category-level analysis further shows that the performance gains of the fine-tuned model stem from stronger domain grounding rather than broader generalization. Improvements concentrate in chemistry, bioprocess engineering, and carbon-management questions (domains that mirror the textbook-derived training corpus), while gains on cross-domain or open-ended items remain limited. Likewise, the largest score increases appear in mechanistic and quantitative reasoning tasks, where the model exhibits a more “textbook-like” reasoning structure: clearer stepwise logic, fewer unsupported mechanisms, and reduced variance in intermediate calculations. These patterns indicate that fine-tuning primarily enhances linguistic precision and structural completeness in domain-specific reasoning, tightening the model's adherence to authoritative textbook logic rather than expanding its general problem-solving breadth.

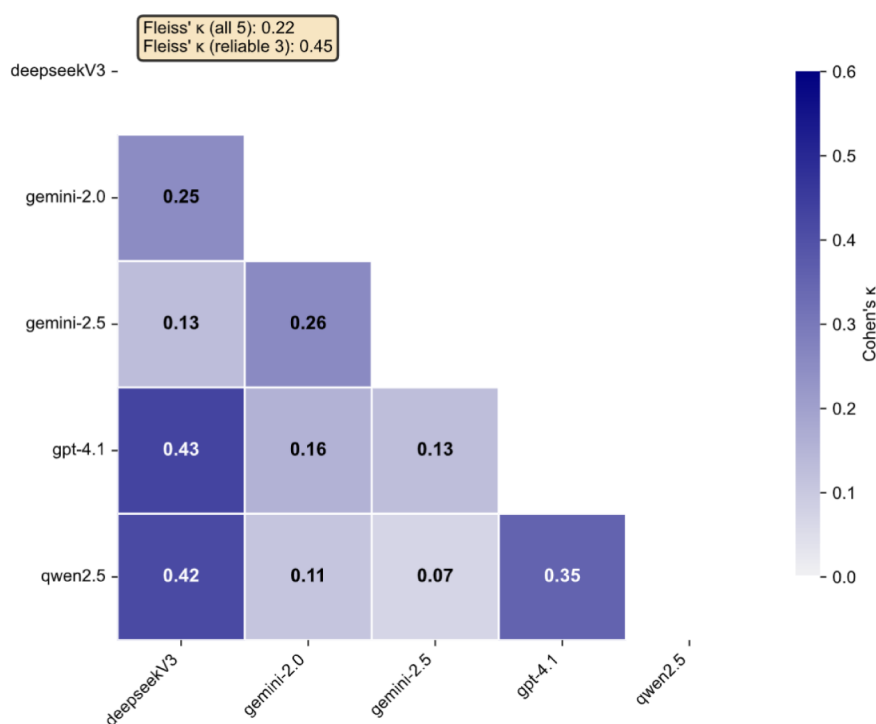


Figure 3. Comparison of interevaluator reliability via Cohen's kappa analysis, where red indicates higher positive agreement and blue indicates negative agreement.

3.2. Agentic Workflows: A Flexible Alternative to Fine-Tuning

Agentic workflows achieve comparable mean performance to supervised fine-tuning while preserving general adaptability. The baseline gpt-4o-mini augmented with agentic tools (agent: 4o-mini) reached a mean score of $80.59 (\pm 14.86)$, closely matching its fine-tuned counterpart gpt-4o-mini-ft (80.61 ± 14.48) but without domain specialization. KDE analysis shows that fine-tuning on the 4o backbone primarily suppresses the low-score tail and increases density near the low-80s, indicating improved stability and fewer severe failures rather than a meaningful shift in peak performance. When the generalist gpt-4.1 mini backbone was integrated into the same workflow, performance increased markedly to $88.40 (\pm 8.54)$, KDE analysis revealing an overall rightward shift, minimal probability mass below ~ 70 , and a visibly thicker high-score region evidence that agentic integration can enhance reasoning consistency and elevate the upper performance range.

In contrast, the fine-tuned agentic workflow (agent: 4o-mini-ft, 77.64 ± 15.84) performed substantially worse. Its KDE distribution formed a narrow peak in the 80–85 range with both tails sharply attenuated, reflecting a conservative answering mode that avoids major errors but rarely achieves top scores. This decline suggests that fine-tuning optimized domain phrasing and deterministic response patterns but reduced flexibility in dynamic reasoning loops. Overall, these results indicate that tool augmentation offers a flexible alternative to fine-tuning, particularly when task diversity and interpretive reasoning are prioritized over marginal accuracy gains.

Analysis of representative error cases (Supporting Information Section 7) shows that the fine-tuned model struggles on mechanistic and multistep problems (e.g., fabricating pathways, omitting transformations, or misapplying basic equations) where the agentic workflow can verify intermediate steps and

correct numerical inconsistencies. This contrast indicates that fine-tuning mainly strengthens domain-specific linguistic precision, while agentic workflows improve procedural adaptability, yielding the largest gains on quantitative, cross-domain, and multistep analytical questions. Consistent with this, KDE distributions show fewer low-score outcomes and tighter dispersion, reflecting enhanced functional robustness rather than increased domain specialization.

3.3. Generalist Models Outperform Specialized Variants

The general-purpose gpt-4.1 mini consistently outperformed all other configurations, achieving the highest overall score of $88.04 (\pm 10.85)$. Its superior performance was most evident in complex, open-ended problems requiring cross-domain reasoning and novel concept integration tasks where smaller or fine-tuned models tended to overfit domain phrasing. When equipped with agentic workflows, its performance increased slightly to $87.51 (\pm 7.69)$, suggesting that tool integration can further refine reasoning consistency and reduce variance without additional training. Generalist gpt-4.1 mini outperforms all other backbone configurations through broad reasoning coverage, achieving the highest scores on analysis, cross-domain synthesis, and evidence-integration tasks, which is also supported by the evidence of its medium effects against others ($p < 0.001$, Cohen's $d = 0.40\text{--}0.67$). Besides, its KDE exhibits a substantially thicker high-score region and minimal low-score mass, reflecting advantages driven by pretraining breadth and architectural efficiency rather than narrow domain optimization. The agentic variant further concentrates probability mass in the 85–95 region, with almost no low-score outcomes.

3.4. Evaluator Consistency and Reliability Analysis

The inter-rater reliability analysis (Figure 3) revealed clear disparities in agreement patterns among the five LLM evaluators. Pairwise Cohen's kappa analysis identified gpt-4.1,

DeepSeek-V3, and Qwen2.5 as a high-consensus group, showing moderate agreement ($\kappa_c = 0.34\text{--}0.41$), with the strongest alignment observed between DeepSeek-V3 and Qwen-2.5 ($\kappa_c = 0.41$). In contrast, the Gemini models exhibited limited consistencies: Gemini-2.5 showed only slight agreement with gpt-4.1 and Qwen2.5 ($\kappa_c = 0.07\text{--}0.13$), whereas Gemini-2.0 maintained similarly low agreement ($\kappa_c < 0.2$) with the high-consensus cohort. Internal agreement between the two Gemini models was also weak ($\kappa_c = 0.26$), approximating chance-level concordance.

When all five evaluators were included, the overall Fleiss' kappa (κ_f) was 0.215, indicating only fair overall agreement. After excluding the Gemini models, the multirater reliability among the remaining three evaluators increased substantially ($\kappa_f = 0.42$), representing moderate consensus. κ_f for all three- and four-evaluator combinations (Supporting Information Section 8) further show that only the DeepSeekV3–gpt-4.1–Qwen2.5 consistently exceeds the reliability threshold.

This outcome shows that the two-stage κ -based screening procedure effectively isolates low-consistency evaluators and maintains the integrity of subsequent comparative analyses. The detection of near-zero-agreement models does not undermine the evaluation framework; rather, it suggests the importance of quantitative reliability prescreening to ensure that benchmarking results are derived from statistically consistent and reproducible judgments.

4. DISCUSSION

This study provides theoretical and practical insights into how large language models can be effectively deployed in environmental domains. The findings reveal that generalist model architecture consistently outperforms domain-specific fine-tuned counterparts in complex, interdisciplinary tasks. Tool-integration capability, rather than linguistic specialization alone, has emerged as a dominant determinant under complex, multistep, and cross-domain tasks. Agentic workflows enable generalist models to overcome the functional rigidity of fine-tuned systems and achieve adaptive, self-correcting reasoning. Evaluator reliability exerts a substantial influence on benchmarking outcomes, underscoring the need for systematic consensus screening. Hybrid deployment strategies that integrate fine-tuning within agentic frameworks anchored in SOTA generalist models offer a balanced pathway toward scalable, interpretable, and resilient environmental AI applications.

4.1. The Performance Superiority of Generalist Architectures

Our results show that while domain-specific fine-tuning enhances precision in narrowly defined tasks (such as pollutant classification,¹⁸ environmental data extraction,⁵⁴ regulatory document review⁵⁵), its benefits diminish rapidly when facing broader environmental challenges. The superior performance of the general-purpose gpt-4.1 mini model in complex, cross-domain scenarios highlight a key characteristic of environmental problems: their intrinsic interdisciplinarity. Environmental issues seldom occur in isolation; they often involve intertwined ecological, social, economic, and technological dimensions. Generalist models, trained on diverse and heterogeneous data, are better equipped to capture these multifaceted interactions.⁵⁶ Their broad contextual knowledge enables them to synthesize information across disciplines, yielding a more integrated and adaptive problem-solving

capability. In contrast, within the scope of our benchmark, fine-tuned models, although strong in specialized domains, did not consistently translate this advantage to tasks requiring cross-domain synthesis. Architectural differences and pretraining diversity may also contribute to the observed trends. These results highlight a potential advantage of generalist architectures in interdisciplinary settings.

The strong performance of gpt-4.1 mini underscores the promise of such generalist architectures, especially when paired with targeted tool augmentation that balances breadth with domain-level precision. Nonetheless, this comparison warrants methodological caution. All models in this study operate within a similar parameter scale ($\sim 7\text{--}8$ billion), but differ in architectural design and optimization strategy. Gpt-4.1 mini integrates extended-context handling, sparse-attention mechanisms, and more efficient parameter allocation, whereas fine-tuned gpt-4o-mini variants employ more compact attention structures and reduced hidden dimensions. These architectural distinctions can influence reasoning depth, training stability, and adaptation under agentic workflows even under comparable parameter counts. Therefore, the observed advantage of the generalist model likely arises from a combination of broader pretraining and architectural efficiency, rather than from scale differences alone. Future research should systematically vary both architecture and model size to disentangle these intertwined factors and establish a clearer understanding of how generalist designs confer robustness in complex environmental applications.

4.2. Tool Integration as a Critical Performance Determinant

The trade-off between model specialization and tool-use flexibility has significant implications for the practical application of LLMs in environmental contexts. Conventional fine-tuning, which relies primarily on textbook-derived analytical reasoning, enhances factual accuracy but does not necessarily endow the model with procedural knowledge needed for dynamic, tool-based reasoning. Without exposure to retrieval, calculation, and error-correction patterns, the fine-tuned backbone may reflect a reduction in flexibility, consistent with recent observations.^{57,58} This limitation reflects that fine-tuning strategies for deployment may need to consider task ecology. In environmental science, decision support often involves integrating quantitative models, spatial data, and policy constraints, that demand continuous interaction with external tools. Thus, future fine-tuning pipelines should couple domain knowledge refinement with tool-use demonstrations and self-correction examples, helping the model better connect domain knowledge with tool-mediated reasoning steps.

From a mechanistic perspective, recent studies⁵⁹ suggest that fine-tuning may induce nonlinear coupling within model parameters, which could reduce representational flexibility when encountering tool-generated input or feedback. This effect further supports the notion that linguistic specialization alone cannot guarantee system-level adaptability. In contrast, generalist models, trained on diverse and interaction-rich corpora, may integrate external computation more effectively.

Viewed through a deployment lens, these findings underscore the need for hybrid adaptation strategies of combining generalist backbones for tool orchestration with fine-tuned components for precision in specific environmental subdomains. Such integrated design may offer a pathway toward

balancing adaptability and accuracy, two properties that are valuable for environmental LLM deployment.

4.3. Agentic Workflows: Functional Advantages over Fine-Tuning

Agentic workflow, which enables dynamic interaction between LLMs and external tools, is a promising and adaptable alternative to conventional fine-tuning in complex environmental contexts. Results show that agentic augmentation allows generalist models to achieve performance comparable to fine-tuned counterparts while maintaining greater flexibility. The baseline gpt-4o-mini model equipped with agentic capabilities reached a mean score of 80.59 (± 14.86), nearly matching the fine-tuned version (80.61 ± 14.48), while the enhanced gpt-4.1 mini configuration achieved 88.04 (± 10.85), surpassing specialized models. These results suggest that task-specific optimization is not always necessary for peak performance. Instead, they suggest the combination of a generalist model and intelligent tool use can achieve remarkable results without the need for task-specific fine-tuning.⁶⁰ This adaptability may be particularly useful in environmental science, where problems often require integrating information from diverse, evolving sources.

Beyond overall accuracy, the closed-loop structure of the agentic workflow, combining task routing, evidence-grounded computation, and self-evaluation, accounts for its improved stability. Each evaluation cycle identifies and corrects logical or numerical inconsistencies, producing progressive refinement and a narrower score distribution. This iterative regulation leads to reduced variance and tighter KDE convergence, indicating that performance gains were accompanied by reduced variance in our benchmark. Dynamic task decomposition further prevents overextension of reasoning chains, while retrieval-based grounding constrains responses within validated environmental knowledge.

Recent advances in adaptive agentic modeling have further highlighted the potential of this paradigm. Emerging agentic systems^{44,45} that integrate reinforcement-style reasoning, physics-informed constraints, and recursive optimization can dynamically incorporate emerging data and theoretical insights without complete retraining. These systems mitigate the rigidity commonly observed in fine-tuned models and support continuous model refinement through iterative reasoning, validation, and feedback mechanisms. Additionally, a spec-driven agentic framework was proposed to bridge expert intent and model execution, facilitating self-consistent updates as new evidence becomes available.⁶¹ Overall, agentic workflows position LLMs as more adaptive analytical tools, capable of maintaining accuracy, interpretability, and contextual responsiveness in rapidly evolving environmental decision domains.

4.4. Evaluator Reliability and Benchmarking Consistency

The observed disparities in interevaluator reliability highlight an important methodological consideration in LLM-based benchmarking. The near-zero agreement observed for certain evaluators (e.g., Gemini models) does not undermine the assessment framework; instead, it reinforces the necessity of systematic reliability screening. Such prescreening ensures that comparative evaluations reflect genuine model consensus rather than noise-driven divergence. The substantial variation in agreement levels, ranging from fair to moderate among the high-consistency group to only slight agreement for the Gemini models, indicates that evaluator choice can meaningfully influence benchmarking outcomes.

These observations have broader implications for the interpretation and generalizability of LLM evaluation studies. When evaluators differ significantly in consistency, conclusions about model performance may become sensitive to the selection of scoring agents, particularly in domains like environmental science where responses may involve nuanced reasoning and multidimensional judgment. In such contexts, relying on low-consistency evaluators can obscure meaningful performance differences or introduce unintended variance.^{62,63}

To mitigate this risk, researchers should adopt rigorous methodological practices to ensure evaluator consensus. This may involve providing evaluators with detailed guidelines and rubrics, conducting training sessions to standardize evaluation procedures, and employing statistical methods to measure and improve interevaluator reliability. Alternatively, researchers may choose to restrict their assessments to established high-reliability models, where evaluator agreement has been demonstrated to be sufficiently high.

4.5. Emergent Multistep Reasoning and Cognitive Implications

Recent advances in reasoning-enhanced large language models provide an additional theoretical perspective on our findings. In particular, frameworks such as PRefLexOR⁶⁴ demonstrate how recursive, preference-based optimization can endow models with the capacity for multistep scientific reasoning. By refining intermediate reasoning paths through reinforcement-learning-style feedback, these systems dynamically balance exploration and convergence, allowing smaller models to exhibit deeper inference and cross-domain adaptability. This mechanism directly complements our interpretation of the specialization–adaptability trade-off: fine-tuning tends to constrain the model's reasoning trajectories via nonlinear parameter coupling, whereas recursive reasoning and chain-of-thought optimization can reintroduce flexibility by encouraging the model to decompose complex environmental tasks into interpretable subproblems. Integrating such multistep reasoning frameworks with agentic workflows therefore represents a promising direction for future work, that could enable self-improving environmental AI systems capable of iterative hypothesis generation, tool-guided exploration, and adaptive knowledge synthesis across domains.

4.6. Deployment Strategies: Integrating Fine-Tuning and Agentic Workflows

This study demonstrates that fine-tuning and agentic workflows are complementary components of a layered environmental AI deployment strategy, rather than competing alternatives. As shown in Table 1, fine-tuning is most appropriate for core, rule-driven tasks where accuracy, consistency, and adherence to established standards are critical, such as regulatory compliance checking, standardized monitoring reports, and routine environmental assessments. In these settings, limited and targeted fine-tuning can effectively encode domain conventions and reduce variance. However, the substantial time and expert labor required for curating textbook-quality data constrain the scalability of fine-tuning and make it less suitable as a primary solution for rapidly evolving decision contexts.

The textbook-based data set introduced here is designed to support such targeted fine-tuning, providing a reusable foundation of shared environmental reasoning and regulatory logic. Without this foundation, fine-tuning on narrow task-specific data risks brittle behavior and poor generalization. In

Table 1. LLMs' Deployment Strategies for Environmental Intelligence

Strategy	Advantages	Disadvantages	Applicable scenarios	Cost
Fine-Tuning	Modest increases in accuracy and stability, while fast response	Less flexible when objectives change	Rule-driven, standardized, fast-response tasks (e.g., compliance checking, pollutant classification)	High (Requires extensive data curation and expert labor)
Agentic Workflows	Flexible, real-time, cross-domain collaboration	Engineering complexity and quality depends on tool chain and prompts	Time-sensitive and data-intensive tasks requiring retrieval, computation, and verification (e.g., risk screening using calculators, data-pipeline tasks of integrating multisource data)	Low-Moderate (Costs depend on tool integration and maintenance by experts)
Model Updates	Strong reasoning ability and adaptability	Lower precision in specific domains	Large-scale, interdisciplinary decision tasks (e.g., scenario planning, cross-domain evidence integration for strategy design)	Low (Faster deployment with fewer data-specific adjustments)

practice, shared cases (e.g., environmental scenarios across multiple Chinese provinces from this train data set) can be incorporated jointly and then combined with region-specific rules to adapt models for local regulatory contexts, reducing downstream curation effort while maintaining consistency. Importantly, the data set construction and validation methodology is itself reusable, allowing the same structured abstraction and expert-in-the-loop process to be applied when generating task-specific or region-specific training data under different environmental decision contexts.

Furthermore, agentic workflows are better built on the most up-to-date, high-capability generalist backbone models, which provide strong cross-domain reasoning and robustness to novel situations. By coupling these models with retrieval, computation, and verification tools, agentic systems can support time-sensitive, data-intensive, and interdisciplinary environmental decisions, such as policy analysis, impact forecasting, and adaptive management, without incurring the high upfront costs of domain-specific retraining. Importantly, fine-tuned models can be selectively incorporated within agentic workflows as verification or checking components, for example to validate regulatory interpretations, numerical outputs, or domain-specific constraints, thereby improving reliability without sacrificing overall system flexibility. While agentic workflows may not always match the precision of fine-tuned models on narrowly defined tasks, their modular integration enables targeted precision where it matters most.

From a practical perspective, these findings support a layered deployment approach: selectively fine-tune models for stable, high-frequency core tasks and for verification roles within agentic systems, while relying on agentic workflows anchored in state-of-the-art foundation models for complex, evolving decision-making. This division of roles aligns domain precision with adaptive intelligence, supporting scalable, transparent, and resilient environmental decision systems.

4.7. Limitations and Future Work

This study has several limitations. First, the data set shows a strong geographic and linguistic bias, with 95% of materials sourced from Chinese-authored textbooks. While this captures China-specific environmental management logic, it limits exposure to global frameworks; future work will incorporate multilingual, cross-regional, and policy-diverse sources to improve generalizability. Second, although the textbook-derived corpus spans multiple environmental subdomains, it masks how individual source types influence model performance. Comparative analyses using disciplinary, policy-oriented, or mixed corpora are needed to disentangle the effects of data diversity from corpus scale. Third, the 328-question test set, though designed to maximize representativeness across subfields and reasoning dimensions, remains moderate in size. Broader and more internationally calibrated benchmarks, with greater expert participation, are required to enhance statistical robustness. Moreover, static Q&A formats cannot fully represent dynamic environmental tasks such as spatial analysis, time-series forecasting, or multicriteria decision-making; future benchmarks should therefore integrate spatiotemporal and scenario-based evaluations. Fourth, while subdomain-specific fine-tuning (e.g., soil, air, water) may improve within-domain accuracy, environmental decision-making often requires cross-domain reasoning. Hierarchical architecture that combines a general environmental foundation with specialized expert modules may better balance adapt-

ability and precision. Fifth, all evaluated models rely on compact backbones (~7–8B parameters), reducing scale-related confounding but limiting examination of capacity effects. Because reasoning performance may correlate with parameter count, future studies should benchmark models across multiple sizes and architectures to isolate the influences of scale, design, and fine-tuning strategy. Finally, methodological and scalability challenges remain. Subtle evaluator biases may persist despite high inter-rater agreement, and the current agentic workflow could benefit from parallelized tool invocation, dynamic load-balance, and expert-rule integration to improve throughput and reliability in production-scale applications.

■ ASSOCIATED CONTENT

SI Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.est.5c09526>.

1. Textbook coverages 2. Training data validation 3. Testing data 4. Evaluation prompt 5. Statistical validation methods and results 6. Representative response comparisons between models 7. Representative error cases and model response analysis 8. Fleiss' kappa for all three- and four-rater combinations (PDF)

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Notes

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■ ABBREVIATIONS

GAI, Generative artificial intelligence; LLMs, Large language models; LCA, Life cycle assessment; RAG, Retrieval-augmented generation; SOTA, State-of-the-art; CI, Confidence intervals; KDE, Kernel density estimation

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